WEEK 7: SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

1. AIM

To implement SQL queries for filtering, sorting, searching, joining, and generating business reports using an E-Commerce Database.

2. SQL QUERIES AND OUTPUTS

A. WHERE Clause

1. Find products with price greater than ₹5000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 5000;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100005
Smartphone
C103         25000              8

P100007
Gaming Monitor
C102         18000              6

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
```

2. Display products available in stock

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Stock_Quantity > 0;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100001
Gaming Headset
C101          2500             10

P100002
Wireless Mouse
C104          1200             25

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P100003
Mechanical Keyboard
C104          3500             15

P100004
Gaming Controller
```

```
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
C102          2800             12

P100005
Smartphone
C103         25000              8

P100006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Bluetooth Speaker
C101          4500             18

P100007
Gaming Monitor
C102         18000              6


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100008
Power Bank
C103          1800             20


8 rows selected.
```

B. ORDER BY Clause

5. Display products from lowest to highest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100002
Wireless Mouse
C104          1200             25

P100008
Power Bank
C103          1800             20

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P100001
Gaming Headset
C101          2500             10

P100004
Gaming Controller
```

```
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
C102          2800             12

P100003
Mechanical Keyboard
C104          3500             15

P100006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Bluetooth Speaker
C101          4500             18

P100007
Gaming Monitor
C102         18000              6


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100005
Smartphone
C103         25000              8


8 rows selected.
```

Find top expensive products

```
SQL> SELECT *
  2  FROM Orders
  3  ORDER BY Order_Date DESC;

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
      1009 C-10001              26-SEP-26         5400 Pending
      1008 C-10001              26-SEP-26         5600 Delivered
      1010 C-10001              26-SEP-26         3600 Delivered
      1007 C-10001              25-SEP-26         3500 Shipped
      1006 C-10001              25-SEP-26         2400 Pending
      1005 C-10001              24-SEP-26         4500 Delivered
      1004 C-10001              23-SEP-26        18000 Pending
      1003 C-10001              22-SEP-26         8400 Delivered
      1002 C-10001              21-SEP-26         3500 Shipped
      1001 C-10001              20-SEP-26         5000 Pending

10 rows selected.
```

C. DISTINCT Clause

9. Display unique product categories

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGOR
-------
C101
C103
C102
C104
```

10. Find different payment methods

```
SQL> SELECT DISTINCT Payment_Method
  2  FROM Payment;

PAYMENT_METHOD
--------------------
Net Banking
Card
Cash
UPI
```

D. Search Products Based on Conditions

11. Find products priced between ₹20,000 and ₹40,000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 20000 AND 40000;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100005
Smartphone
C103         25000              8
```

Retrieve Customer and Product Information

16. Display customer details with their orders

```
SQL> SELECT *
  2  FROM Customer C
  3  JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                             PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------
C-10001              Ricky                Ponting
ricky200@edss.com               9876543210 14-FEB-23       1001
C-10001              20-SEP-26         5000 Pending

C-10001              Ricky                Ponting
ricky200@edss.com               9876543210 14-FEB-23       1002
C-10001              21-SEP-26         3500 Shipped

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                             PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------

C-10001              Ricky                Ponting
ricky200@edss.com               9876543210 14-FEB-23       1003
C-10001              22-SEP-26         8400 Delivered

C-10001              Ricky                Ponting
ricky200@edss.com               9876543210 14-FEB-23       1004

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                             PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------
C-10001              23-SEP-26        18000 Pending
```



```
-------------------- -------------------- --------------------
EMAIL                          PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------
ricky200@edss.com              9876543210 14-FEB-23       1006
C-10001              25-SEP-26         2400 Pending

C-10001              Ricky                Ponting
ricky200@edss.com              9876543210 14-FEB-23       1007
C-10001              25-SEP-26         3500 Shipped


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------
C-10001              Ricky                Ponting
ricky200@edss.com              9876543210 14-FEB-23       1008
C-10001              26-SEP-26         5600 Delivered

C-10001              Ricky                Ponting
ricky200@edss.com              9876543210 14-FEB-23       1009
C-10001              26-SEP-26         5400 Pending

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CREATED_A   ORDER_ID
------------------------------ ---------- --------- ----------
CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- --------- ------------ --------------------

C-10001              Ricky                Ponting
ricky200@edss.com              9876543210 14-FEB-23       1010
C-10001              26-SEP-26         3600 Delivered


10 rows selected.
```

Display product details with category information

```
SQL> SELECT *
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
P100001
Gaming Headset
C101          2500             10 C101

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
Electronics
Electronic products
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
Accessories
Computer accessories


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
P100003
Mechanical Keyboard
C104          3500             15 C104

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
P100004
Gaming Controller
C102          2800             12 C102

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
Gaming
Gaming products


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY CATEGOR
------- ---------- -------------- -------
CATEGORY_NAME
```

Display products purchased by each customer

```
SQL> SELECT C.Customer_ID, P.Product_Name
  2  FROM Customer C
  3  JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID
  5  JOIN ORDER_DETAILS I
  6  ON O.Order_ID = I.Order_ID
  7  JOIN Product P
  8  ON I.Product_ID = P.Product_ID;

CUSTOMER_ID
--------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
C-10001
Gaming Headset

C-10001
Mechanical Keyboard

C-10001
Gaming Controller


CUSTOMER_ID
--------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
C-10001
Gaming Monitor

C-10001
Bluetooth Speaker

C-10001
Wireless Mouse
```

```
CUSTOMER_ID
--------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
C-10001
Mechanical Keyboard

C-10001
Gaming Controller

C-10001
Gaming Controller


CUSTOMER_ID
--------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
C-10001
Power Bank


10 rows selected.
```

F. Multiple Filtering Conditions

20. Find electronics products costing more than ₹10,000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 'C102'
  4  AND Price > 10000;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100007
Gaming Monitor
C102         18000              6
```

Search products containing the word “Mobile”

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Product_Name LIKE '%Mobile%';
```

Output: No rows selected.

Business Reports

23. Product Availability Report

```
SQL> SELECT Product_Name,
  2         Price,
  3         Stock_Quantity,
  4         CASE
  5           WHEN Stock_Quantity > 0 THEN 'Available'
  6           ELSE 'Out of Stock'
  7         END AS Availability_Status
  8  FROM Product;

PRODUCT_NAME
--------------------------------------------------------------------------------
----
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Gaming Headset
      2500             10 Available

Wireless Mouse
      1200             25 Available

Mechanical Keyboard
      3500             15 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
----
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Gaming Controller
      2800             12 Available

Smartphone
     25000              8 Available

Bluetooth Speaker
      4500             18 Available


PRODUCT_NAME
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
----
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Gaming Monitor
     18000              6 Available

Power Bank
      1800             20 Available


8 rows selected.
```

Total Number of Customers

```
SQL> SELECT COUNT(*) AS Total_Customers
  2  FROM Customer;

TOTAL_CUSTOMERS
---------------
              1

SQL> SELECT *
  2  FROM Customer
  3  WHERE Created_At >= TRUNC(SYSDATE, 'MM');
```

Total Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
          10
```

Completed Orders

```
SQL> SELECT COUNT(*) AS Completed_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Completed';

COMPLETED_ORDERS
----------------
               0
```

Pending Orders

```
SQL> SELECT COUNT(*) AS Pending_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Pending';

PENDING_ORDERS
--------------
             4

SQL> SELECT COUNT(*) AS Cancelled_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Gaming Headset
C101          2500             10

P100008
Power Bank
C103          1800             20


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100002
Wireless Mouse
C104          1200             25


8 rows selected.
```

Most-Reviewed Products

```
SQL> SELECT Product_ID, COUNT(*) AS Total_Reviews
  2  FROM Product_Review
  3  GROUP BY Product_ID
  4  ORDER BY Total_Reviews DESC;

PRODUCT TOTAL_REVIEWS
------- -------------
P100001             1
P100004             1
P100002             1
P100003             1
```

Display Available Products

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Stock_Quantity > 0;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100001
Gaming Headset
C101          2500             10

P100002
Wireless Mouse
C104          1200             25

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P100003
Mechanical Keyboard
C104          3500             15

P100004
Gaming Controller
```

```
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
C102          2800             12

P100005
Smartphone
C103         25000              8

P100006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Bluetooth Speaker
C101          4500             18

P100007
Gaming Monitor
C102         18000              6


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P100008
Power Bank
C103          1800             20


8 rows selected.
```

3. RESULT

The SQL queries were executed using SQL*Plus on the E-Commerce Database. Filtering, sorting, searching, joining, and business reporting operations were performed. Some queries returned no matching records, and the rating query could not be executed because the Rating column was not available in the Product table.

4. CONCLUSION

Thus, the required SQL operations were performed on the E-Commerce Database using SQL*Plus. The practical demonstrated the use of WHERE, ORDER BY, DISTINCT, BETWEEN, LIKE, JOIN, COUNT, GROUP BY, and CASE statements.